



Cambridge IGCSE™

BIOLOGY

Paper 1 Multiple Choice (Core)

0610/13

May/June 2026

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

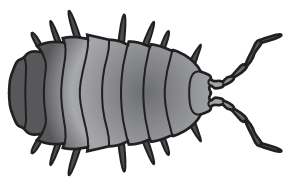
- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

- 1 The diagram shows a species of arthropod.



Which group does this arthropod belong to?

- A arachnid
 - B crustacean
 - C insect
 - D myriapod
- 2 How many pairs of jointed legs does an insect have?
- A two pairs
 - B three pairs
 - C four pairs
 - D more than six pairs
- 3 Which type of cells are specialised to move mucus?
- A ciliated cells
 - B palisade mesophyll cells
 - C root hair cells
 - D white blood cells

- 4 Which structures are present in bacterial cells?

key

✓ = present in bacterial cells

✗ = **not** present in bacterial cells

	cell membrane	cell wall	chloroplast
A	✓	✓	✗
B	✓	✗	✗
C	✗	✓	✓
D	✗	✗	✓

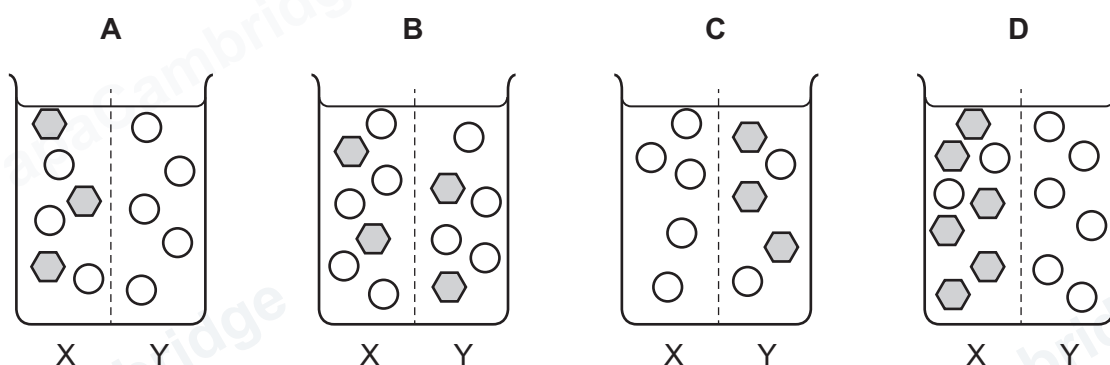
- 5 An image of a spider is 275 mm long.

The magnification of the image is $\times 25$.

What is the actual size of the spider?

- A** 0.09 mm **B** 11.0 mm **C** 300 mm **D** 6875 mm
- 6 The diagrams show beakers with different concentrations of glucose solution separated by a permeable membrane. The membrane allows both glucose molecules and water molecules to move through.

In which diagram will glucose molecules diffuse from X to Y the fastest?



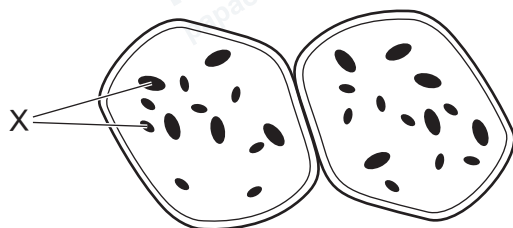
key

⬡ glucose molecules

○ water molecules

- 7 The diagram shows cells from a storage organ of a plant.

The structures, X, are stained blue-black by iodine solution.



What do the structures, X, contain?

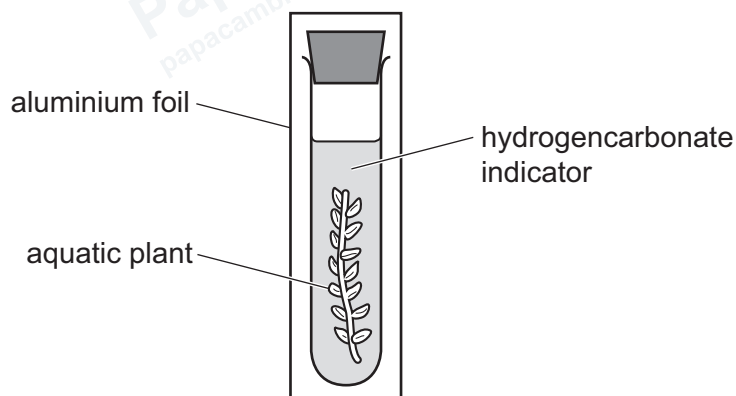
- A chlorophyll
- B fat
- C starch
- D sugar

- 8 Which statements describe a catalyst?

- 1 a substance that decreases the rate of a chemical reaction
- 2 a substance that increases the rate of a chemical reaction
- 3 a substance that is changed by the chemical reaction
- 4 a substance that is **not** changed by the chemical reaction

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

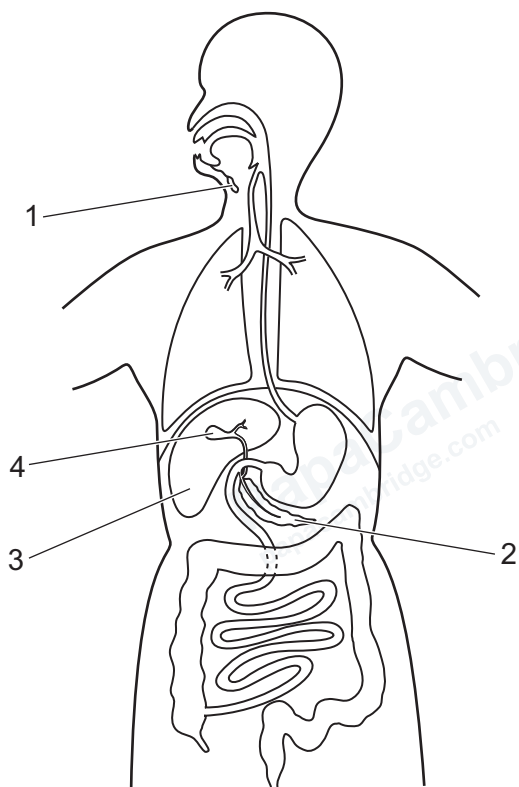
- 9 The diagram shows apparatus used to investigate gas exchange in an aquatic plant.



Which colour would the hydrogencarbonate indicator be after several hours?

- A blue
 - B purple
 - C red
 - D yellow
- 10 Which carbohydrate is used to build plant cell walls?
- A cellulose
 - B glycogen
 - C starch
 - D sucrose
- 11 Which structure in a leaf is adapted for photosynthesis by being near the upper surface and containing a large number of chloroplasts?
- A epidermis cell
 - B guard cell
 - C palisade mesophyll cell
 - D xylem vessel

12 The diagram shows the human digestive system.



Which row identifies the organs labelled 1–4?

	1	2	3	4
A	gall bladder	liver	pancreas	salivary glands
B	gall bladder	liver	salivary glands	pancreas
C	salivary glands	pancreas	gall bladder	liver
D	salivary glands	pancreas	liver	gall bladder

13 Which enzymes can digest oily fish?

key

✓ = yes

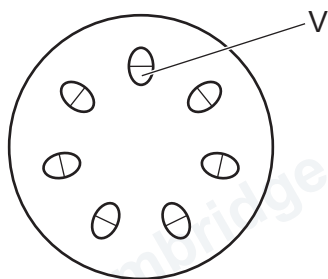
X = no

	amylase	lipase	protease
A	✓	✓	X
B	✓	X	✓
C	X	✓	✓
D	X	X	✓

14 Which part of the alimentary canal is where most water is absorbed into the blood?

- A** oesophagus
- B** colon
- C** stomach
- D** small intestine

15 The diagram shows a cross-section of a plant stem.



What is the function of the tissue labelled V?

- A** transporting dissolved nutrients and mineral ions
- B** transporting dissolved nutrients only
- C** transporting water and mineral ions
- D** transporting water only

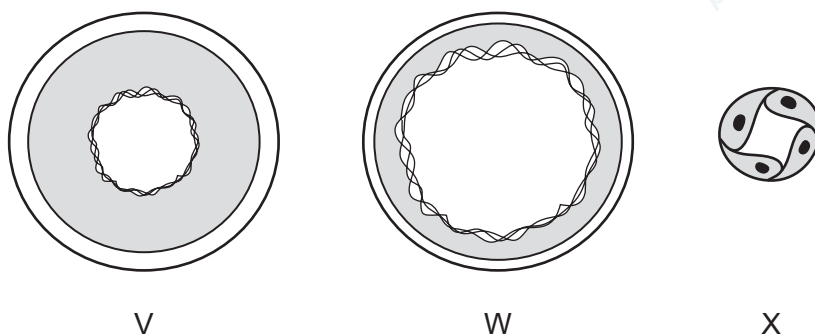
16 Which conditions will increase the rate of transpiration in a well-watered plant the most?

- A high temperature and high wind speed
- B high temperature and low wind speed
- C low temperature and high wind speed
- D low temperature and low wind speed

17 Which structure in the human heart separates the left and right ventricles?

- A atria
- B coronary artery
- C one-way valve
- D septum

18 The diagram shows cross-sections through three types of blood vessel, **not** drawn to the same scale.



Which section is from a vein and which section is from a capillary?

	vein	capillary
A	V	W
B	W	V
C	W	X
D	X	W

- 19 Which row shows how the concentrations of carbon dioxide and oxygen change in the blood as it passes through the lungs?

	carbon dioxide	oxygen
A	increase	decrease
B	increase	increase
C	decrease	decrease
D	decrease	increase

- 20 What are ways in which energy from respiration is used in living organisms?

key

✓ = energy is used for this

✗ = energy is **not** used for this

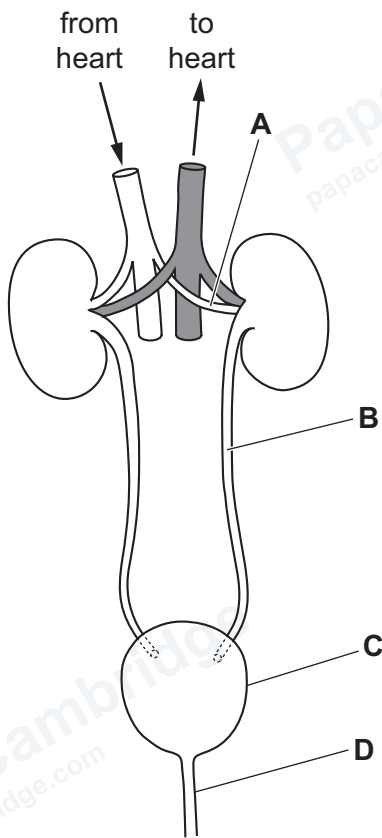
	active transport	diffusion	osmosis	protein synthesis
A	✓	✓	✓	✗
B	✓	✗	✗	✓
C	✓	✗	✓	✓
D	✗	✓	✗	✓

- 21 Which cells produce carbon dioxide during anaerobic respiration?

	muscle cells	yeast cells
A	no	no
B	no	yes
C	yes	no
D	yes	yes

22 The diagram shows structures associated with the human urinary system.

Which structure is the urethra?



23 Which statements about the mammalian nervous system are correct?

- 1 A synapse is a junction between two neurones.
- 2 Relay neurones pass electrical impulses to receptors.
- 3 The central nervous system (CNS) consists of the brain only.
- 4 The peripheral nervous system (PNS) consists of nerves outside the brain and spinal cord.

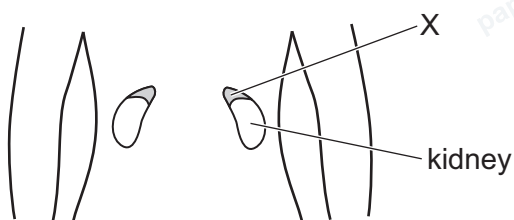
A 1 and 2

B 1 and 4

C 2 and 4

D 3 and 4

- 24 The diagram shows the position of the kidneys in the body.



Which hormone is produced by the gland labelled X?

- A adrenaline
 - B insulin
 - C oestrogen
 - D testosterone
- 25 A seed is planted under the soil and the shoot grows upwards.
- What is the name of this response when the shoot is growing under the soil?
- A gravitropism
 - B phototropism
 - C photosynthesis
 - D transpiration
- 26 What is defined as a substance that is taken into the body and modifies or affects chemical reactions in the body?
- A antibody
 - B drug
 - C hormone
 - D pathogen
- 27 What is fertilisation?
- A a sperm entering an egg cell
 - B a sperm nucleus fusing with an egg cell nucleus
 - C the division of a zygote
 - D the movement of a sperm towards an egg cell

28 Which structure produces pollen grains?

- A anther
- B filament
- C stigma
- D style

29 Methods of disease transmission are listed.

- 1 blood transferred from an infected person
- 2 holding hands with an infected person
- 3 sharing a drinking cup with an infected person
- 4 unprotected sexual intercourse with an infected person

By which methods of disease transmission can the human immunodeficiency virus (HIV) be transmitted between people?

- A 1 and 2 B 1 and 4 C 2 and 3 D 3 and 4

30 What is defined as a length of DNA that codes for a protein?

- A amino acid
- B chromosome
- C gene
- D nucleus

31 A female fruit fly with short wings mates with a male fruit fly with long wings. All the offspring are long-winged.

Two of these offspring mate with each other.

Which percentage of their offspring will have long wings?

- A 25% B 50% C 75% D 100%

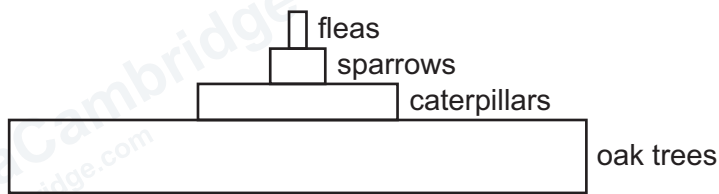
32 Four characteristics are listed.

- ABO blood group in humans
- body mass in humans
- seed shape in peas
- seed colour in peas

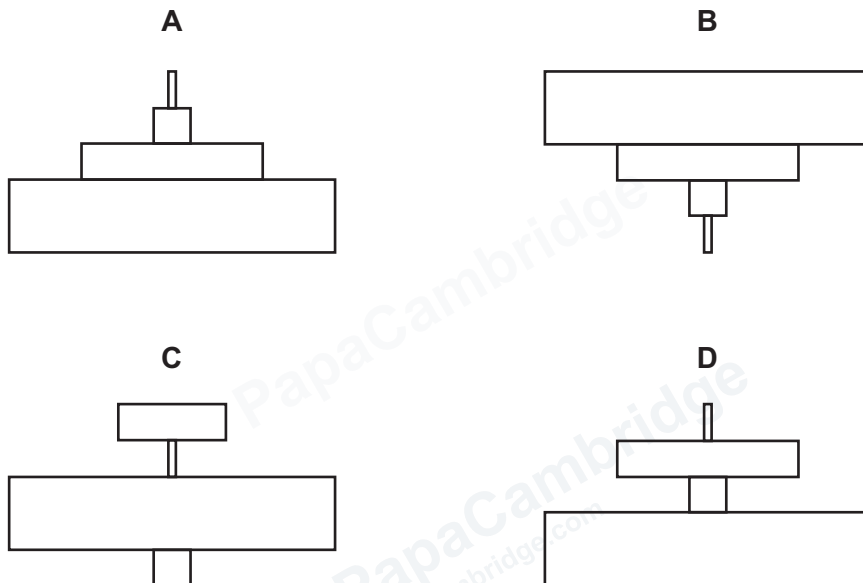
How many of these characteristics show continuous variation?

- A** one **B** two **C** three **D** four

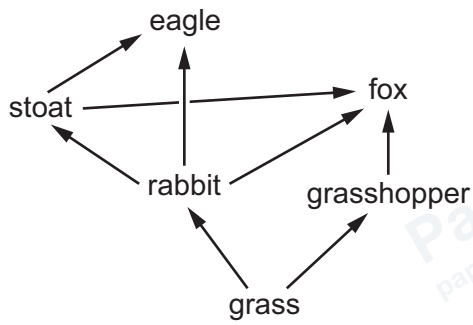
33 The diagram shows a pyramid of biomass for a food chain in a woodland ecosystem.



What is the pyramid of numbers for this food chain?



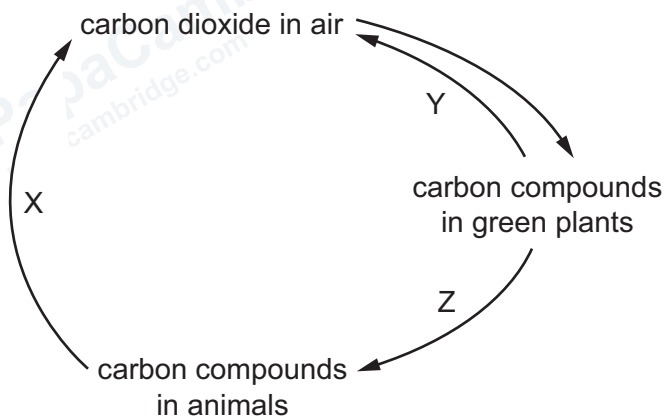
34 The diagram shows a food web.



Which statement about this food web is correct?

- A It has five carnivores.
- B It has five consumers.
- C It has one herbivore.
- D It has two producers.

35 The diagram represents part of the carbon cycle.

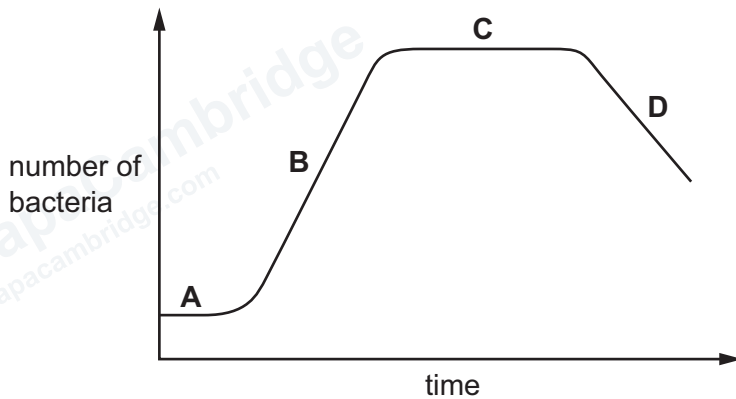


Which processes are represented by X, Y and Z?

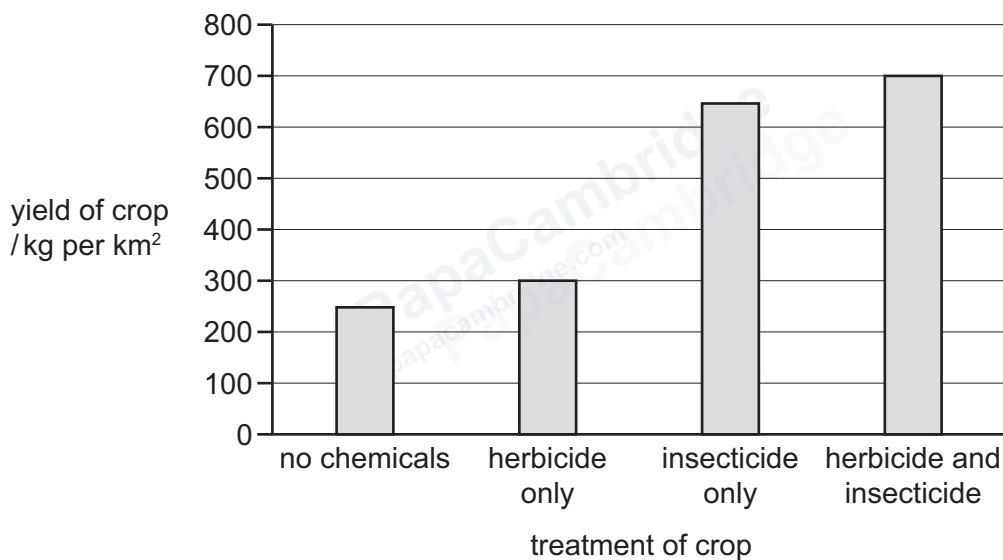
	X	Y	Z
A	photosynthesis	feeding	combustion
B	respiration	respiration	feeding
C	combustion	feeding	photosynthesis
D	feeding	respiration	combustion

36 The graph shows a population growth curve for bacteria.

Which letter identifies the lag phase on the graph?



37 The graph shows the yields of a crop when treated with different combinations of a herbicide and an insecticide.



Which conclusions can be made about the treatment of the crop with the herbicide and the insecticide?

- 1 Adding the herbicide and the insecticide together increases the crop yield the most.
- 2 Adding the insecticide only has a bigger effect on crop yield than adding the herbicide only.
- 3 Adding the insecticide must kill all insect pests that consume the crop.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

38 What is **least** likely to result from deforestation?

- A increase in flooding
- B increase in species
- C loss of habitats
- D loss of soil

39 Which row correctly shows if each gas is responsible for the enhanced greenhouse effect?

	carbon dioxide	methane
A	no	no
B	no	yes
C	yes	no
D	yes	yes

40 What is an example of genetic modification?

- A inserting a gene for human insulin production into bacteria
- B selective breeding to produce cattle with high milk yields
- C using herbicides to kill unwanted plants
- D using insecticides to kill insect pests

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